Longitudinal Analysis of Orthodontic Growth

Prepared for:
Internal Use

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A statwise Application

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Analysis Overview

This analysis examines changes in craniofacial distance over time using the Orthodont dataset from the nlme R package. The dataset includes 108 measurements from 27 children (16 boys and 11 girls), each assessed at ages 8, 10, 12, and 14 years.

We used linear mixed-effects models (LMMs) to account for repeated measurements within children and address two main research questions:

- Does craniofacial distance increase with age?
- Do boys and girls differ in their rates of change over time?

We compared random-intercept and random-intercept-plus-slope models to select an appropriate random-effects structure. We then evaluated the age × sex interaction to assess differences in growth rates between boys and girls.

Results include model comparisons, estimated annual changes with 95% confidence intervals, model-based predicted trajectories, and diagnostic assessments.

Age Effect on Craniofacial Distance

We used linear mixed-effects models to evaluate changes in craniofacial distance with age. Since each child was measured repeatedly, a random intercept was included to account for differences between children and within-child correlation.

To determine whether children also differed in their growth rates, we compared two models:

- **Random Intercept Only:** Allows each child to have a different baseline measurement but assumes the same growth rate.
- **Random Intercept + Random Slope:** Allows each child to have a different baseline measurement and growth rate.

The models were compared using a likelihood ratio test (LRT).

Table 1: Comparison of Random-Effects Structures

Model	AIC	BIC	LRT Chi-square	df	p-value
Random Intercept Only	451.39	462.12	—	—	
Random Intercept + Slope	451.21	467.30	4.18	2	0.124

Adding a random slope did not significantly improve model fit (LRT: $\chi^2(2) = 4.18$, $p = 0.124$). Therefore, the simpler random-intercept model was selected.

The estimated annual change in craniofacial distance was 0.66 mm/year.

Table 2: Estimated Annual Change in Craniofacial Distance

Outcome	Annual Change (mm/year)	Lower 95% CI	Upper 95% CI	p-value
Craniofacial Distance	0.660	0.538	0.783	<0.001

Model diagnostics were generally acceptable, with some deviations from normality in the distribution tails. All observations were retained.

Sex Differences in Craniofacial Growth

We evaluated whether boys and girls had different rates of change in craniofacial distance over time.

Two linear mixed-effects models with random intercepts were fitted:

- **Without Interaction:** Includes age and sex, assuming the same growth rate for boys and girls.
- **With Interaction:** Includes age, sex, and the age × sex interaction, allowing different growth rates for boys and girls.

We compared the models using a likelihood ratio test (LRT).

Table 3: Comparison of Models With and Without Age × Sex Interaction

Model	AIC	BIC	LRT Chi-square	df	p-value
Without Interaction	444.86	458.27	—	—	
With Interaction	440.64	456.73	6.22	1	0.013

Adding the age × sex interaction significantly improved model fit (LRT: $\chi^2(1) = 6.22$, $p = 0.013$). Therefore, the interaction model was selected.

Estimated Annual Growth Rates by Sex

We estimated the annual change in craniofacial distance for boys and girls using the selected model.

Table 4: Estimated Annual Growth Rates by Sex

Sex	Annual Change (mm/year)	Lower 95% CI	Upper 95% CI	p-value
Male	0.784	0.630	0.939	<0.001
Female	0.480	0.294	0.665	<0.001

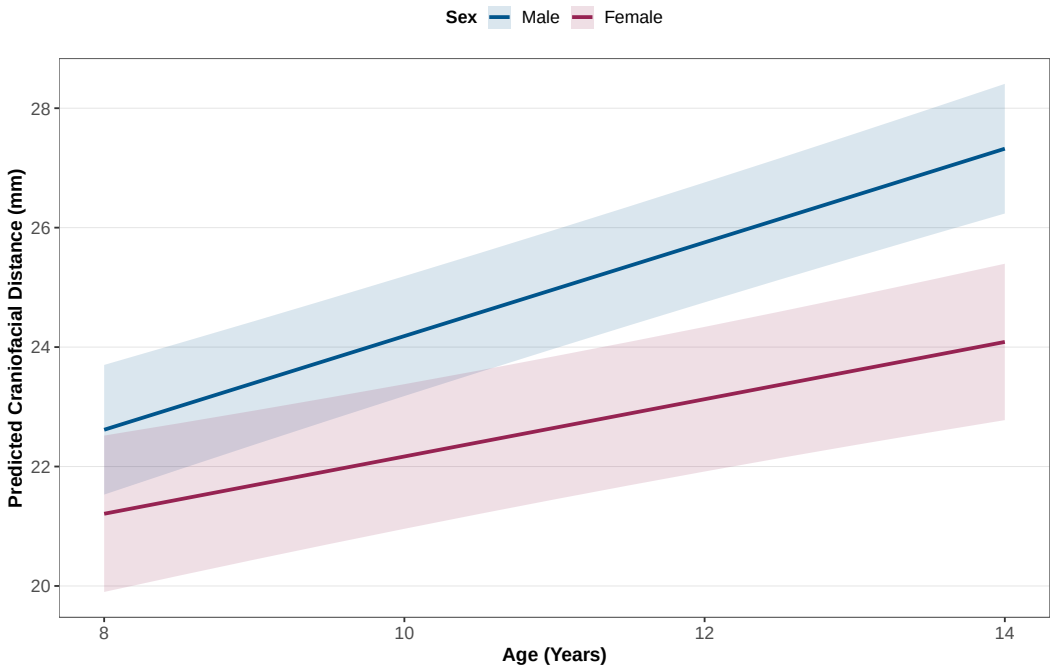
Craniofacial distance increased with age in both groups. The estimated annual increase was 0.784 mm/year for boys and 0.48 mm/year for girls. Boys showed a greater annual increase than girls.

Model-Based Predicted Trajectories

We visualized the predicted mean craniofacial distance by age and sex using the selected interaction model. Shaded areas represent 95% confidence intervals.

Model-based predicted trajectories show that craniofacial distance increases with age in both boys and girls. Boys have higher predicted measurements and a steeper growth trajectory, consistent with the significant age × sex interaction. The difference between the predicted trajectories increases with age.

Figure 1: Model-Based Craniofacial Growth Trajectories by Sex



Model Diagnostics

Model diagnostics indicated generally acceptable fit. Residuals showed no clear systematic pattern or substantial heteroscedasticity. Q-Q plots suggested approximate normality of residuals and random intercepts, with some deviations in the tails.

Figure 2: Residuals vs. Fitted

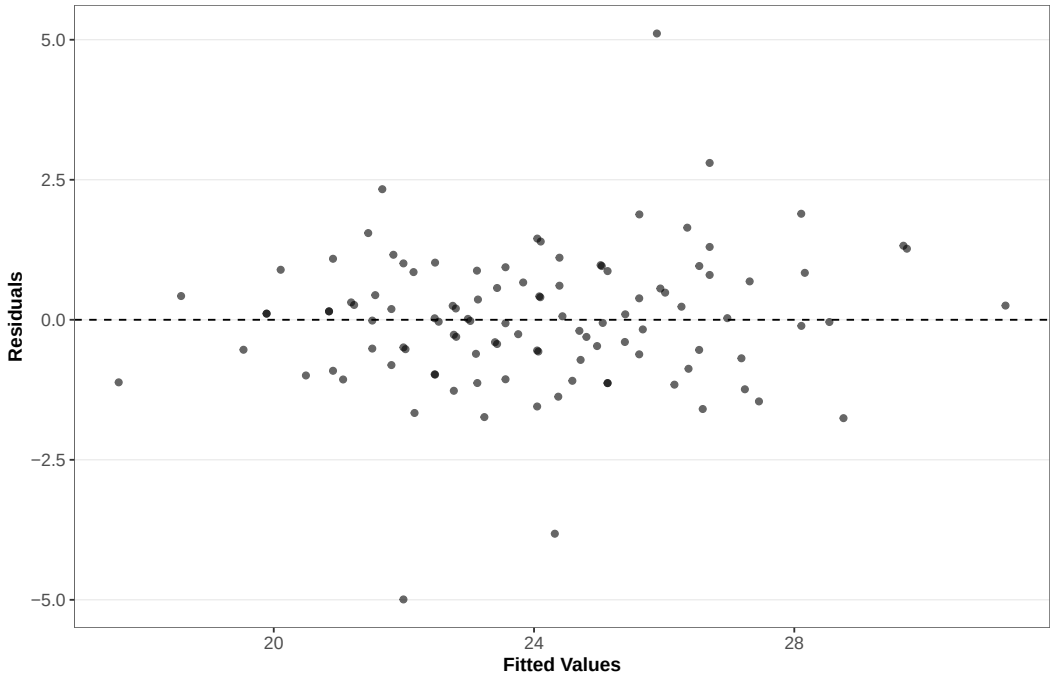


Figure 3: Normal Q-Q Plot: Residuals

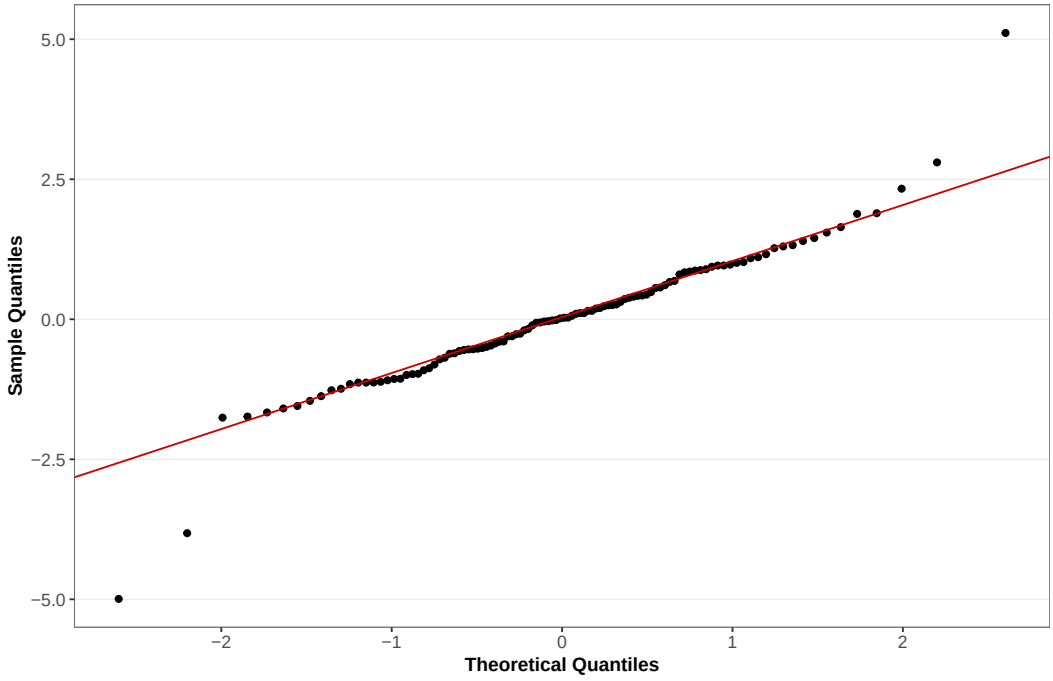
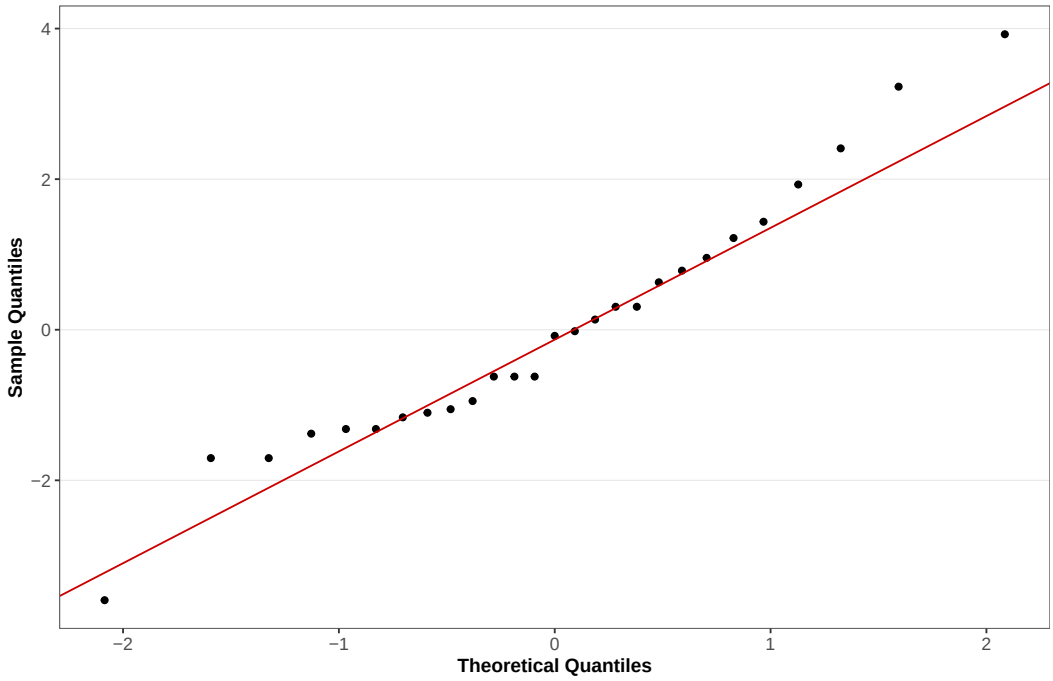


Figure 4: Normal Q-Q Plot: Random Intercepts



Summary

Craniofacial distance increased with age in both boys and girls. The random-intercept-only model was selected because adding a random slope did not significantly improve model fit.

The age × sex interaction was statistically significant (p = 0.013), indicating that growth rates differed between boys and girls. The estimated annual increase was 0.784 mm/year (95% CI: 0.630–0.939) for boys and 0.480 mm/year (95% CI: 0.294–0.665) for girls.

Model-based predicted trajectories showed a steeper increase in craniofacial distance among boys. Model diagnostics indicated generally acceptable fit, with some deviations from normality in the distribution tails.

Overall, the analysis demonstrates how linear mixed-effects models can be used to evaluate longitudinal changes, account for repeated measurements, and compare growth trajectories between groups.